
ST-SIMDIFF: BALANCING SPATIOTEMPORAL SIMILARITY AND DIFFERENCE FOR EFFICIENT VIDEO UNDERSTANDING WITH MLLMs

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ABSTRACT

When Multimodal Large Language Models (MLLMs) ingest long videos, the visual token sequence expands with both temporal duration and spatial resolution; because the self-attention that consumes these tokens scales quadratically with sequence length, the induced computational and memory burden rapidly becomes the dominant obstacle to long-form and interactive deployment. Prevailing efficiency methods attenuate this cost by pruning or merging tokens according to importance or similarity, yet in doing so they equate efficiency with the suppression of redundancy alone. Two consequences follow. First, they overlook a complementary and equally informative dimension of video content—the changes and turning points through which a video’s narrative actually unfolds. Second, they model spatial and temporal relations in isolation rather than jointly, and therefore capture only part of the redundancy that a video contains. We reorganize the problem around a single principle: similarity is the signal for identifying redundancy, whereas difference is the signal for preserving key events. Instantiating this principle, we introduce ST-SimDiff, a training-free token selection framework that assembles the visual tokens into a unified spatio-temporal graph—within which spatial and temporal associations are represented jointly—and then executes two complementary selection procedures in parallel. The first, a similarity-driven procedure, performs community detection over the graph to retain a compact set of representative tokens that encode stable, redundant content; the second, a temporal-difference-driven procedure, localizes points of abrupt feature change along the time axis and preserves the tokens that mark critical dynamic transitions. Their union constitutes a minimal token subset that at once represents static content and captures the events that define the plot. Extensive experiments across multiple MLLM architectures and three long-form video understanding benchmarks show that ST-SimDiff consistently outperforms state-of-the-art compression methods while substantially reducing inference time and memory, and that at moderate token budgets it recovers—and on several benchmarks exceeds—the accuracy of the full-token model. Our code will be released upon acceptance.

1 INTRODUCTION

The rapid maturation of Large Language Models (LLMs) has driven a correspondingly rapid advance in Large Vision-Language Models (LVLMs), which now exhibit strong image and video understanding capabilities (Zhang et al., 2024c; Liu et al., 2024c). To process video, contemporary LVLMs sample a clip into a sequence of frames and encode each frame into hundreds or even thousands of visual tokens before forwarding them to the LLM (Bai et al., 2025; Li et al., 2024). Effective as this paradigm is, the visual token count grows explosively with video duration and resolution; and because the self-attention that consumes these tokens scales quadratically with sequence length, the induced computational and memory burden quickly becomes prohibitive, sharply curtailing the use of LVLMs for long-video analysis and real-time interaction (Fu et al., 2024b). Reducing the visual token budget without sacrificing accuracy is therefore a central prerequisite for scaling video understanding, and it is the problem this work directly addresses.

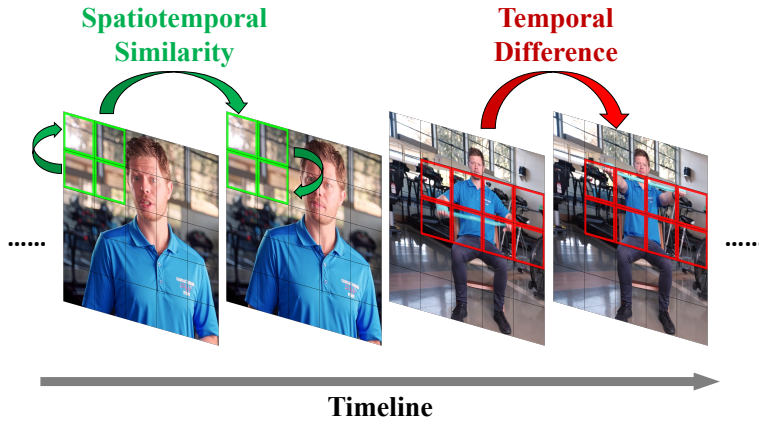


Figure 1: The core motivation of ST-SimDiff. Efficient long-video understanding requires addressing two complementary regimes simultaneously. Spatiotemporal similarity (left) is exploited to compress redundant information, both spatially within a frame and temporally across adjacent frames, whereas temporal difference (right) is detected to preserve the key actions and events that define the plot. Attending only to the former, as prior methods do, discards the latter.

Existing efforts to improve LVLM efficiency fall into two broad families, both of which reduce cost by suppressing what is redundant. The first is importance-based pruning: FastV (Chen et al., 2024) observes that the attention an LVLM allocates to visual information is redundant across layers and, on that basis, prunes the visual tokens that receive low attention scores in the deeper layers. The second is similarity-based merging and selection, which exploits the observation that visual tokens are highly similar either across adjacent frames or within a single frame; FrameFusion (Fu et al., 2024b) merges similar tokens from adjacent frames, whereas VisionZip (Yang et al., 2025) selects dominant tokens directly at the vision-encoder level. Productive as these two families are, both frame token reduction narrowly as the elimination of redundancy.

Despite their effectiveness at reducing visual tokens, these methods share two fundamental limitations. First, they treat correlation one axis at a time—either spatial correlation within a frame or temporal correlation at fixed positions across frames—rather than modeling spatio-temporal association jointly; this disjoint treatment leaves them unable to characterize the complex dynamics that span both dimensions, and consequently caps the redundancy they can exploit. Second, and more consequentially, they optimize exclusively for informational commonality—whether formalized as similarity or as importance—and therefore inherit a shared blind spot: they discount the very changes and differences that carry meaning. Because a video’s narrative is driven by turning events, a scheme that seeks only similarity will smooth over a sudden action or scene transition, discarding precisely the tokens most responsible for correct interpretation. Efficiency and fidelity are thus placed in conflict exactly where the content is most informative.

We resolve this conflict by adopting a single organizing principle: similarity is the signal for identifying redundancy, while difference is the signal for capturing key moments. An effective video token compression algorithm must accordingly satisfy two objectives jointly—representing a video’s stable content with the fewest possible tokens, and precisely preserving its key changes—rather than trading one against the other. Guided by this principle, we introduce **ST-SimDiff**, a dual token selection framework built upon a spatio-temporal graph, whose two components operate in parallel on a shared representation. The first, **similarity-based representative token selection**, treats every visual token as a node and connects nodes by feature similarity along both the spatial and the temporal dimension; a graph community detection algorithm then isolates tightly connected token clusters, each corresponding to a stable, persistent visual element, and retains from each cluster only the few most central tokens as its representatives. The second, **difference-based event token selection**, inspects the edges oriented along the time axis: when the similarity between corresponding tokens in adjacent frames drops sharply, the transition is identified as a turning point and the associated tokens are marked as critical event tokens to be retained. Passing the union of the two token sets

to the LLM yields a subset that is at once compact and faithful to both the static and the dynamic structure of the video. Our contributions can be summarized as follows:

- **A reframing of token reduction.** We argue that token similarity and token difference deserve equal standing in LVLM efficiency research, recasting compression as the joint task of suppressing redundancy *and* preserving change rather than as the suppression of redundancy alone—a reorientation that, to our knowledge, prior token-compression work has not made explicit.
- **A unified framework that instantiates the principle.** We construct a spatio-temporal graph that models the spatial and temporal relationships among video tokens within a single representation, and on top of it we design a dual token selection strategy that concurrently identifies representative tokens and pivotal event tokens. The procedure is training-free and adds only linear overhead relative to the LVLM’s quadratic attention, which makes it straightforward to integrate into existing video LVLM pipelines.
- **Consistent empirical gains across models and budgets.** Across three long-form video benchmarks and two distinct base models—with a third architecture examined in the appendix—ST-SimDiff substantially reduces the visual token budget while maintaining or improving accuracy, consistently outperforming state-of-the-art compression methods and, at moderate budgets, matching or exceeding the full-token model.

2 RELATED WORK

2.1 LARGE VISION LANGUAGE MODELS

The rapid advancement of Large Language Models (LLMs) (Touvron et al., 2023; Achiam et al., 2023; Ouyang et al., 2022) has propelled a parallel wave of progress in multimodal understanding, giving rise to a family of increasingly capable Large Vision Language Models (LVLMs) (Liu et al., 2024b; Bai et al., 2023). These models encode images or video frames into visual tokens through pre-trained visual encoders and forward the tokens to an LLM, bridging the two modalities with an alignment module—typically an MLP (Liu et al., 2024a;b) or a Q-Former (Li et al., 2023; Dai et al., 2023). This design confers strong performance across a wide range of multimodal tasks (Xi et al., 2025; Holdaway et al., 2024; Zhang et al., 2025a), and in video understanding it has proven especially promising for longer, higher-resolution, and more complex videos (Bai et al., 2025; Li et al., 2024; Zhang et al., 2024c; Liu et al., 2024c).

This capability, however, carries a pronounced computational cost that is most acute in the video regime. Because a video is processed as a continuous sequence of frames, its visual token count far exceeds that of a single image and grows steeply with duration, straining both the training and the inference efficiency of LVLMs. Processing this large body of video tokens efficiently while curbing computational overhead is therefore a central open problem in contemporary LVLM research, and the one this work directly targets.

2.2 VISUAL TOKEN COMPRESSION

Motivated by the substantial overhead of video processing, visual token compression has emerged as a principal lever for improving LVLM efficiency. Existing approaches attack the problem along two complementary axes: token importance and token similarity.

Early work centered on importance-based pruning, which identifies and discards the visual tokens that contribute least to model performance, quantifying token importance through attention scores or related feature metrics; FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b) are representative. Subsequent methods (Shang et al., 2024; Ye et al., 2025) additionally account for token similarity, recognizing that even high-importance tokens may be mutually redundant and can be consolidated without appreciable information loss. In this vein, FrameFusion (Fu et al., 2024b) couples similarity-based merging with importance-based pruning, VisionZip (Yang et al., 2025) selects dominant tokens while merging contextual information, and VisPruner (Zhang et al., 2025b) exploits visual cues to remove redundant items.

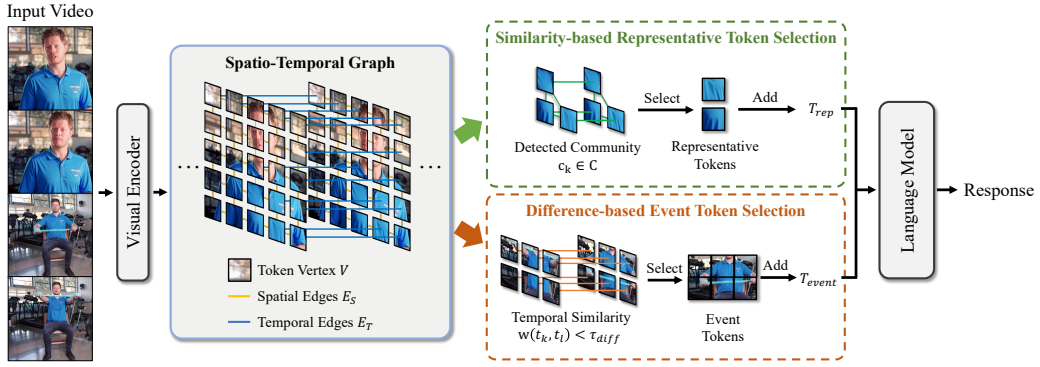


Figure 2: Overview of the ST-SimDiff framework. A single spatio-temporal graph, constructed from the video’s visual tokens, feeds two parallel selection procedures: Similarity-based Representative Token Selection, which compresses redundant static content via community detection, and Difference-based Event Token Selection, which preserves the tokens that mark abrupt temporal transitions. The union of the two selections constitutes the reduced token set, which is then reconciled to the target budget.

These advances notwithstanding, prior work leaves two gaps that jointly motivate our design. First, it seldom models the global spatio-temporal similarity structure among tokens: video redundancy manifests both spatially and temporally, yet similarity-driven methods such as FrameFusion attend chiefly to the temporal similarity between adjacent frames and thereby forgo the richer joint spatio-temporal relations—an omission that caps how thoroughly they can exploit inherent redundancy and that erodes performance under aggressive token reduction. Second, the literature as a whole optimizes for commonality—importance or similarity—and provides no mechanism for actively preserving *difference*, the abrupt changes that constitute a video’s key events. ST-SimDiff is designed to close both gaps simultaneously.

3 PROBLEM DEFINITION

Given a video V and a text query Q , a standard Large Vision-Language Model (LVLM) first encodes the video into a full sequence of N visual tokens, $T_{\text{full}} = \{t_1, t_2, \dots, t_N\}$. Feeding this sequence to the LLM is computationally demanding because the self-attention mechanism scales quadratically with the token count, incurring $O(N^2)$ cost. We therefore cast token reduction as the design of a selection function $f(\cdot)$ that maps the full sequence T_{full} and a target compression ratio r to a substantially smaller subset $T_{\text{sub}} = f(T_{\text{full}}, r)$, with the objective of maximizing the LVLM’s downstream task performance on this subset subject to the budget constraint that its cardinality $|T_{\text{sub}}|$ be approximately $r \cdot N$. The central difficulty is accordingly to construct $f(\cdot)$ such that the retained subset T_{sub} concentrates the information most critical to the task within the allotted budget.

4 ST-SIMDIFF

4.1 METHOD OVERVIEW

As illustrated in Figure 2, ST-SimDiff comprises two components that operate in parallel on a single spatio-temporal graph constructed from the video’s visual tokens: Similarity-based Representative Token Selection (SRTS) and Difference-based Event Token Selection (DETS). This shared graph is precisely what enables the two components to address redundancy and change on a common footing, rather than as two disconnected pipelines. SRTS condenses the video’s stable, redundant content: community detection over the graph groups highly similar tokens into clusters corresponding to persistent visual elements such as static backgrounds, and from each cluster it retains only a few maximally central tokens, forming the representative set T_{rep} . DETS, complementarily, captures the video’s crucial dynamic shifts: it inspects the temporal edges of the graph and localizes the moments at which the similarity between corresponding tokens in adjacent frames drops sharply; these high-

difference points denote turning points, and the tokens framing each transition are preserved as critical event tokens, forming the set T_{trans} . Combining the two selections yields the final subset $T_{\text{sub}} = T_{\text{rep}} \cup T_{\text{trans}}$, which is at once highly compact and information-rich: it retains stable content while precisely capturing the events that drive the video forward.

4.2 SPATIO-TEMPORAL GRAPH CONSTRUCTION

Given a video, the vision encoder transforms it into a sequence of tokens $T = \{t_1, t_2, \dots, t_N\}$. Each token $t_k \in T$ occupies a specific spatio-temporal location and carries a feature vector $\mathbf{x}_k$; we index that location by its frame position $T(t_k)$, its spatial height index $H(t_k)$, and its spatial width index $W(t_k)$.

We encode the relationships among these tokens as a spatio-temporal graph $G = (V, E)$ whose vertex set V comprises all tokens in T . The edge set decomposes into two disjoint families, $E = E_S \cup E_T$, capturing spatial and temporal adjacency respectively; representing both within a single graph is what permits the subsequent selection to reason over spatial and temporal structure jointly. The spatial edges E_S connect tokens that are spatially adjacent within the same frame:

$$E_S = \{(v_i, v_j) \in V \times V \mid T(t_i) = T(t_j) \text{ and } |H(t_i) - H(t_j)| + |W(t_i) - W(t_j)| = 1\} \quad (1)$$

The temporal edges E_T connect tokens that occupy the same spatial position in adjacent frames, thereby encoding temporal continuity along the time axis:

$$E_T = \{(v_i, v_j) \in V \times V \mid H(t_i) = H(t_j), W(t_i) = W(t_j), \text{ and } |T(t_i) - T(t_j)| = 1\} \quad (2)$$

Each edge $w(v_i, v_j) \in E$ is weighted by the cosine similarity between the feature vectors of its endpoints, so that the edge weight directly quantifies the semantic affinity of the two tokens it connects:

$$w(v_i, v_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

Restricting edges to spatial and temporal neighbors keeps the graph sparse—each token connects to only a small, constant number of neighbors—while still encoding local spatial relationships and temporal continuity; this sparse construction is what keeps the framework’s overhead linear and furnishes the unified foundation on which our dual-path selection strategy operates.

4.3 SIMILARITY-BASED REPRESENTATIVE TOKEN SELECTION

Video content is pervaded by spatio-temporal redundancy. A static scene that persists for several seconds yields background tokens that remain highly similar over time, and an object traversing the frame produces tokens that, although their spatial positions shift, retain strong semantic correlation. Because prior methods treat temporal and spatial similarity in isolation, they capture only part of this redundancy; our spatio-temporal graph, by contrast, exposes both forms jointly. Within the graph, semantically related tokens coalesce into dense “communities” or “clusters” according to feature similarity, irrespective of the particular frame or location in which they arise, so that selecting a small number of representatives from each community becomes a highly efficient mechanism for compressing redundant information while preserving core semantics.

To isolate strongly correlated clusters, we first sparsify the graph by thresholding: fixing a similarity threshold τ_{sim} , we retain only those edges whose weight exceeds it, yielding the pruned graph $G' = (V, E')$ with $E' = \{(v_i, v_j) \in E \mid w(v_i, v_j) > \tau_{\text{sim}}\}$. This step severs weak associations and leaves communities well separated. We then apply a graph community detection algorithm (e.g., the Louvain method (Blondel et al., 2008)) to G' to recover the token clusters $C = \{c_1, c_2, \dots, c_m\}$. Within each community $c_k \in C$ we rank tokens by centrality, defining the centrality score $S_c(t_a)$ of a token $t_a \in c_k$ as its mean similarity to all other tokens in the community:

$$S_c(t_a) = \frac{1}{|c_k| - 1} \sum_{t_b \in c_k, b \neq a} w(t_a, t_b) \quad (4)$$

We then set the intra-community retention rate to the globally specified compression ratio r and preserve, from each community c_k , its top $\lceil |c_k| \cdot r \rceil$ tokens by centrality. Applying the same proportional rule to every community handles all cases uniformly, including single-node communities of unique tokens, for which the lone node is retained by construction. Formally, the representative set T_{rep} is defined as

$$T_{\text{rep}} = \bigcup_{k=1}^m \text{TopK}_{t \in c_k, \text{score} = S_c(t)}(|c_k| \cdot r) \quad (5)$$

where the TopK operator returns the k highest-scoring elements of a given set. By retaining the most central tokens of each semantic cluster, SRTS compresses the static and persistent content of the video while safeguarding the semantics that each cluster represents.

4.4 DIFFERENCE-BASED EVENT TOKEN SELECTION

If similarity defines the “norm” of a video, difference defines its “events”. A model attuned only to similarity may excel at describing “what is” yet falter on “what happened,” because a video’s plot is driven by events whose essence is change—the appearance of a new object, the onset of an action, a scene transition. In our formulation, such changes surface as a sharp drop in the feature similarity of temporally adjacent tokens; accurately localizing these “turning points” of large difference is therefore indispensable both for understanding dynamic content and for correctly answering “when” and “why” questions. This is exactly the signal that similarity-only methods discard, and the one DETS is designed to recover.

Because difference along the temporal dimension so reliably signals a key event—the entry of a new object, a scene change, or the onset or termination of an action all produce abrupt shifts in the visual features at corresponding positions across adjacent frames—DETS operates specifically on the temporal edges E_T of the spatio-temporal graph. For a temporal edge $(v_k, v_l) \in E_T$ connecting the temporally adjacent tokens t_k and t_l , we compare its similarity against a difference threshold τ_{diff} (e.g., set from the tail of the temporal-edge difference distribution): whenever the similarity falls below the threshold, i.e., $w(t_k, t_l) < \tau_{\text{diff}}$, the transition is deemed significant and the subsequent token t_l is retained as a critical event token. Formally, the event set T_{event} is defined as

$$T_{\text{event}} = \{t_l \mid \exists t_k \text{ s.t. } (v_k, v_l) \in E_T, T(t_l) > T(t_k), \text{ and } w(v_k, v_l) < \tau_{\text{diff}}\} \quad (6)$$

This criterion selectively preserves the tokens that mark moments of significant temporal change, ensuring that the crucial dynamic aspects of the video’s narrative—precisely the content that redundancy-driven compression is prone to discard—survive the reduction process.

4.5 OVERALL REDUCTION PROCESS

ST-SimDiff computes the representative set T_{rep} and the event set T_{event} in parallel and takes their union, $T_{\text{candidate}} = T_{\text{rep}} \cup T_{\text{event}}$, as the initial candidate set; because the two procedures share the same underlying graph, the budget is not partitioned a priori between them but is instead allocated implicitly, in proportion to how much stable content and how much abrupt change the video actually contains. To meet the target token count $N_{\text{target}} = \lceil r \cdot N \rceil$ exactly, we append a final reconciliation step: whenever $|T_{\text{candidate}}|$ exceeds N_{target} , we discard the $|T_{\text{candidate}}| - N_{\text{target}}$ tokens of lowest importance, where, following Chen et al. (2024), a token’s importance is its attention score in the shallow layers of the LLM. Applying this single importance criterion across the union treats representative and event tokens on equal terms, so that whichever tokens the two procedures agree are least informative are the first to be removed, and the method conforms to any prescribed computational budget while retaining the information both selection procedures identified as critical.

The overall method thus composes two complementary mechanisms: graph-based token selection and attention-based dynamic pruning. The former exploits the intrinsic structure of video content—both its similarity and its difference—to guarantee that core information is retained, whereas the latter supplies a flexible control that pins the output to any target budget. Together they render the compression process at once principled and adaptive, striking a robust balance between efficiency and performance.

4.6 COMPUTATIONAL COMPLEXITY ANALYSIS

We now establish that the overhead of ST-SimDiff is linear in the number of tokens by analyzing its three stages—Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection (SRTS), and Difference-based Event Token Selection (DETS)—in turn.

Graph construction computes, for each of the N visual tokens, the cosine similarity to its spatial and temporal neighbors. Because every token has only a small, constant number of neighbors, a single pass over all tokens suffices, giving complexity $O(Nd)$, where d is the token feature dimension.

SRTS then recovers the token clusters. As detailed in our implementation (Section 5.2), we employ a connected-components algorithm whose near-linear complexity is $O(N + |E|)$; because the edge count satisfies $|E| \leq 3N$, this reduces to $O(N)$. The subsequent per-community filtering step could in principle become expensive—a naive all-pairs similarity computation within a community c_k would cost $O(|C| \cdot |c_k|^2 \cdot d)$ —so we cap the maximum community size and, in the rare event that a community exceeds a predefined threshold (e.g., $|c_k| > \sqrt{N}$), partition it; this bound keeps filtering efficient and prevents it from dominating the overall cost. Although mainstream alternatives such as the Louvain and Leiden methods yield more expressive community structure, their complexity approaches $O(N \log N)$; we therefore adopt the simpler connected-components algorithm to hold processing overhead to a minimum, a trade-off whose accuracy cost we find negligible in practice.

DETS makes a single pass over the temporal edges, comparing each edge’s similarity against the threshold to flag significant changes, and thus likewise runs in linear time $O(Nd)$.

Taken together, these linear-time stages give ST-SimDiff an overall complexity of $O(Nd)$, asymptotically far below the $O(N^2d)$ cost of the LLM’s self-attention. The framework consequently delivers substantial token reduction at a computational footprint that is negligible relative to the model’s main inference cost.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines To evaluate ST-SimDiff comprehensively and to probe its generality across architectures, we adopt two distinct base models, LLaVA-Video (Zhang et al., 2024c) and NVILA (Liu et al., 2024c), and benchmark our approach against a representative range of state-of-the-art video token compression techniques. The baselines span the two dominant strategies: **importance-based reduction methods**, namely FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b), and **hybrid reduction methods** that combine importance with similarity, namely VisionZip (Yang et al., 2025), FrameFusion (Fu et al., 2024b), and PruMerge (Shang et al., 2024). To ensure a fair comparison, every method is evaluated under the identical token retain ratio, frame count, video sampling, prompt template, and decoding configuration; the only variable across rows of Tables 1 and 2 is the token selection rule itself. We further bracket every setting with the original uncompressed model (Vanilla) as an upper bound and random sampling of an equal number of tokens as a lower bound.

Benchmarks We assess long-video understanding on three challenging and complementary benchmarks—VideoMME (Fu et al., 2024a), LongVideoBench (Wu et al., 2025), and EgoSchema (Mangalam et al., 2024)—selected so as to probe distinct facets of the task. VideoMME is a comprehensive benchmark for foundational video understanding that spans diverse video types across three duration categories: Short, Medium, and Long. LongVideoBench targets fine-grained retrieval and reasoning over details in long videos through a “referring reasoning” task and covers four video lengths (15s, 60s, 600s, and 3600s). EgoSchema is a question-answering dataset of long-form, first-person videos that stresses a model’s grasp of character intentions and of the causal chains linking actions. Taken together, the three benchmarks exercise general perception, long-range detail reasoning, and egocentric causal understanding, and thus span the principal competencies demanded of long-video LVLMs.

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
LLaVA-Video	-	-	-	63.3	-	-	58.2	57.3
Token Retain Ratio $r = 30\%$								
FastV	68.2	58.6	51.2	59.3	48.0	59.9	53.5	51.3
PruMerge	70.2	57.3	52.2	59.9	49.6	60.5	54.7	50.9
FasterVLM	71.3	57.6	51.4	60.1	51.0	61.3	55.8	52.6
VisionZip	68.4	55.9	50.4	58.3	46.9	60.3	53.2	53.0
FrameFusion	74.0	59.8	50.0	61.3	49.7	63.3	56.0	53.0
ST-SimDiff (Ours)	74.7	61.9	53.0	63.2	52.5	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$								
FastV	73.9	61.4	51.3	62.2	50.1	62.2	55.7	54.7
PruMerge	72.3	58.7	52.8	61.3	51.4	63.2	56.9	54.6
FasterVLM	74.0	59.2	51.8	61.7	51.3	62.2	56.4	56.2
VisionZip	72.6	57.7	51.6	60.6	51.3	63.2	56.8	54.2
FrameFusion	74.6	61.2	52.0	62.6	51.7	64.2	57.6	55.8
ST-SimDiff (Ours)	76.2	62.4	52.7	63.8	52.3	64.3	57.9	57.3

Table 1: Performance comparison against state-of-the-art token compression methods on the LLaVA-Video-7B base model with 64 input frames (%), reported on VideoMME (Short/Medium-/Long/Overall), LongVideoBench (Relation/Perception/Overall), and EgoSchema under token retain ratios of $r = 30\%$ and $r = 50\%$. All methods share identical frame counts, sampling, and decoding settings; the full-token model (Upper Bound) is listed for reference, and the best result among all compression methods in each column is shown in **bold**. ST-SimDiff attains the best overall score in every setting, and at $r = 50\%$ its VideoMME accuracy (63.8) surpasses the full-token upper bound (63.3).

5.2 IMPLEMENTATION DETAILS

All experiments run on NVIDIA L20 GPUs (48 GB VRAM) under Ubuntu 22.04, with inference evaluated through the *lmms-eval* (Zhang et al., 2024a) library. For LLaVA-Video we retain its original configuration of 64 input frames, and to hold computational conditions comparable across architectures we likewise fix NVILA to 64 input frames. We set the similarity threshold to $\tau_{\text{sim}} = 0.8$ and the difference threshold to $\tau_{\text{diff}} = 0.2$ —values justified by the sensitivity analysis in the appendix, where performance is shown to be stable across a broad surrounding range and the same optimum to transfer across benchmarks and base models—and evaluate at token compression ratios of $r = 30\%$ and $r = 50\%$. For community detection we adopt connected-component finding, which our complexity analysis identifies as the most favorable point in the trade-off between processing time and accuracy.

5.3 COMPARISON WITH STATE-OF-THE-ARTS

We evaluate ST-SimDiff against a range of mainstream efficient video-language-model (VLM) compression methods on the three long-form video understanding benchmarks—VideoMME, LongVideoBench, and EgoSchema. To assess both generalization across architectures and robustness across budgets, we report detailed results on two base models, LLaVA-Video-7B and NVILA-8B, at token retain ratios of 30% and 50%. The results appear in Table 1 and Table 2.

(1) *ST-SimDiff leads consistently across models, budgets, and benchmarks.* It attains the best overall score on both VideoMME and LongVideoBench in all four (base model $\times$ ratio) settings, and

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
NVILA-Video	-	-	-	61.5	-	-	56.3	52.9
Token Retain Ratio $r = 30\%$								
FastV	69.4	55.3	48.8	57.9	50.8	55.6	53.0	49.7
PruMerge	71.0	53.7	49.9	58.2	49.3	58.1	53.4	47.5
FasterVLM	72.6	56.9	51.0	60.1	48.7	57.8	53.0	49.3
VisionZip	71.4	56.2	49.7	59.1	46.3	56.2	50.9	48.9
FrameFusion	72.1	55.7	48.7	58.8	51.3	59.3	54.9	51.3
ST-SimDiff (Ours)	73.3	57.7	49.7	60.2	51.7	59.2	55.2	51.7
Token Retain Ratio $r = 50\%$								
FastV	71.9	58.3	49.3	58.9	49.5	58.8	53.9	50.2
PruMerge	71.2	54.8	46.8	57.6	49.7	58.7	53.9	48.9
FasterVLM	74.6	57.8	50.1	60.8	48.5	58.1	53.0	50.5
VisionZip	72.8	58.1	50.6	60.5	50.4	58.9	54.4	50.3
FrameFusion	72.7	57.2	48.3	59.4	51.0	59.2	54.8	52.6
ST-SimDiff (Ours)	73.9	59.7	51.4	61.7	52.3	61.3	56.5	52.5

Table 2: Performance comparison against state-of-the-art token compression methods on the NVILA-Video-8B base model with 64 input frames (%), reported on VideoMME, LongVideoBench, and EgoSchema under token retain ratios of $r = 30\%$ and $r = 50\%$, under the same evaluation protocol as Table 1. The full-token model (Upper Bound) is listed for reference, and the best result among all compression methods in each column is shown in **bold**. That ST-SimDiff leads on VideoMME and LongVideoBench across both ratios on this second, independently developed architecture corroborates the framework’s generality.

the best EgoSchema score in three of the four; the advantage is therefore systematic rather than incidental. The margin is widest on EgoSchema at $r = 30\%$, where, on LLaVA-Video, ST-SimDiff reaches 56.0 against the strongest baseline’s 53.0—a 3.0-point gain—and it carries over to the harder Medium and Long splits of VideoMME as well. On the less aggressive $r = 50\%$ budget the margins over the strongest baseline narrow, as expected once every method operates well within the token budget, yet ST-SimDiff still leads: most notably, at $r = 50\%$ its overall accuracy matches or surpasses the full-token model on several benchmarks, reaching 63.8 on VideoMME with half the tokens against the full-token 63.3 and equalling the full-token EgoSchema score of 57.3 on LLaVA-Video. That a training-free reduction to half the tokens can recover, and in places exceed, full-token accuracy is a meaningful outcome for long-video inference, and it motivates a closer look at where the baselines fall short.

(2) *The baselines’ limitations follow from what they optimize for.* Among the importance-based baselines (FastV, FasterVLM), FasterVLM is generally the stronger, yet both handle temporal redundancy inefficiently: by retaining tokens that are individually important but mutually repetitive, they cap their achievable compression efficiency. The hybrid baselines (PruMerge, VisionZip, FrameFusion) are more competitive—FrameFusion is typically the best among them—because they additionally exploit similarity. Even so, every baseline optimizes for representative “commonality” alone and therefore shares a structural blind spot: it can overlook the key narrative information carried by “turning points” and “abrupt events.”

(3) *The two design choices account for the observed gains.* Collectively the baselines trace a progressively finer understanding of video “redundancy,” but they remain limited on two fronts: they lack a synergistic treatment of the joint spatio-temporal similarity among visual tokens, and so capture complex redundancy only incompletely; and they omit “difference” detection, and so fail to

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	60.3	56.2	54.8
+ Sim	Spatial	61.5	56.5
	Temporal	61.7	56.8
	Spa. + Tem.	62.6	57.0
++ Diff (Proposed)	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$			
Baseline	63.2	56.7	56.5
+ Sim	Spatial	63.3	57.0
	Temporal	63.7	57.2
	Spa. + Tem.	63.7	57.8
++ Diff (Proposed)	63.8	57.9	57.3

Table 3: Component ablation on LLaVA-Video-7B (%), reported on VideoMME, LongVideoBench, and EgoSchema at token retain ratios of $r = 30\%$ and $r = 50\%$. Starting from an importance-only Baseline, we progressively add the similarity module (+Sim, in its spatial-only, temporal-only, and joint spatio-temporal variants) and then the difference module (++Diff), which completes the framework; the best result in each column is in **bold**. Accuracy improves monotonically as each component is added, and the joint spatio-temporal variant dominates both single-axis variants—evidence that the two modules are complementary rather than redundant.

actively preserve the changes that define plot development. ST-SimDiff addresses both directly—modeling spatio-temporal relationships within a unified graph and introducing explicit difference detection—and the consistent margins across Tables 1 and 2 are the empirical consequence.

5.4 ABLATION STUDY

To isolate the contribution of each core component, we conduct a controlled ablation that begins from a Baseline using only fundamental importance-based pruning and then progressively adds the similarity selection module (+Sim) and the difference selection module (++Diff). All variants are evaluated at retain ratios of 30% and 50%; the results are reported in Table 5.

(1) *Joint spatio-temporal similarity outperforms either axis alone.* Adding the +Sim module to the Baseline improves performance consistently across all three benchmarks and both ratios. Decomposing the similarity strategy into spatial-only (Spatial), temporal-only (Temporal), and joint spatio-temporal (Spa. + Tem.) variants reveals a clear ordering, with the joint variant uniformly the strongest—at $r = 30\%$, for instance, it reaches 62.6 on VideoMME against 61.5 (spatial) and 61.7 (temporal)—which confirms that modeling spatio-temporal association jointly, rather than one axis at a time, is the more effective route to redundancy compression.

(2) *Difference detection contributes most where tokens are scarce.* Building on the joint similarity module, adding the difference module (++Diff) completes the framework and yields a further gain, most pronounced under the aggressive $r = 30\%$ budget (e.g., VideoMME rises from 62.6 to 63.2 and EgoSchema from 55.3 to 56.0). This pattern substantiates the module’s complementary role: whereas +Sim aggressively compresses redundancy, ++Diff acts as a safeguard that identifies and preserves the unique event tokens defining the video’s narrative. The increment is smaller at $r = 50\%$, as anticipated, since under a looser budget +Sim already retains many of these event tokens and overall accuracy approaches its upper bound—an observation that precisely locates the value of difference detection in the demanding high-compression regime. Together the two modules realize our central thesis: it is the balance between similarity-based redundancy compression and difference-based event preservation, rather than either mechanism alone, that makes the framework effective, and most conspicuously so when tokens are scarce.

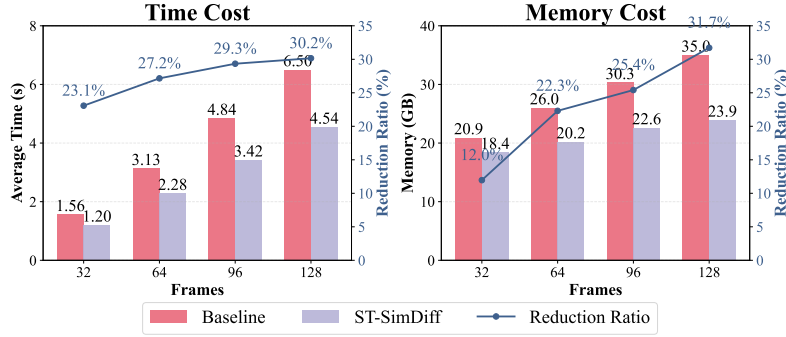


Figure 3: Computational cost of ST-SimDiff versus the uncompressed LLaVA-Video baseline, measured as average inference time and peak GPU memory usage under a 30% token budget across inputs of 32 to 128 frames. Both the time and the memory savings widen as the video lengthens—exactly the regime in which efficiency is most critical.

5.5 COMPUTATIONAL COST ANALYSIS

To quantify the practical efficiency of ST-SimDiff, we measure its inference time and peak GPU memory under a 30% token budget and compare against the uncompressed baseline (Baseline LLaVA-Video) over input lengths ranging from 32 to 128 frames. The results, shown in Figure 3, translate our linear-time complexity analysis into concrete wall-clock and memory savings, and confirm that the overhead of graph construction and token selection is more than offset by the shortened LLM inference it enables.

Inference Time ST-SimDiff yields a substantial and length-dependent speedup. The baseline’s average inference time climbs steeply with the frame count, from 1.56 seconds at 32 frames to 6.50 seconds at 128 frames, whereas our method curbs this growth to just 4.54 seconds at 128 frames. Decisively, the time-saving rate itself widens with video length, rising from 23.0% at 32 frames to 30.2% at 128 frames, so that the advantage compounds precisely in the long-video regime where efficiency matters most.

Memory Cost The memory picture is equally favorable. The baseline’s peak GPU memory grows linearly with video length, from 20.9 GB at 32 frames to 35.0 GB at 128 frames; by sharply reducing the number of tokens forwarded to the LLM, ST-SimDiff holds peak memory to 23.9 GB for 128-frame videos—a 31.7% reduction relative to the baseline. This headroom allows longer videos to be processed on memory-constrained hardware, thereby broadening the range of settings in which the model can be deployed.

In sum, ST-SimDiff couples leading task accuracy with pronounced gains in both time and memory efficiency; that these savings widen with video length underscores its suitability for long-form video understanding, the setting for which it is designed.

6 CONCLUSION

This paper confronts the computational inefficiency of Multimodal Large Language Models on long videos—a problem rooted jointly in the excessive number of visual tokens and in the neglect of critical content changes—and resolves it through a single organizing principle: similarity identifies redundancy, whereas difference captures key events. We instantiate this principle in ST-SimDiff, a training-free framework that assembles the visual tokens into a spatio-temporal graph modeling their spatial and temporal associations and then applies a dual selection strategy in which similarity-based selection compresses redundancy through community detection and temporal-difference-based selection preserves the tokens at key transition points. Across three long-form video benchmarks and two distinct base models—and, in the appendix, a third architecture—ST-SimDiff substantially reduces inference time and memory while consistently surpassing state-of-the-art compression methods and, at moderate budgets, matching or exceeding the full-token model. By elevating transitional

events to equal standing with content representativeness, this work charts a direction for efficient video representation and, more broadly, for how token reduction in MLLMs might be reconceived. A natural avenue for future work is to make the selection query-aware, so that content globally redundant yet salient to a specific question can also be retained.

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A VISUALIZATION OF THE TOKEN SELECTION PROCESS

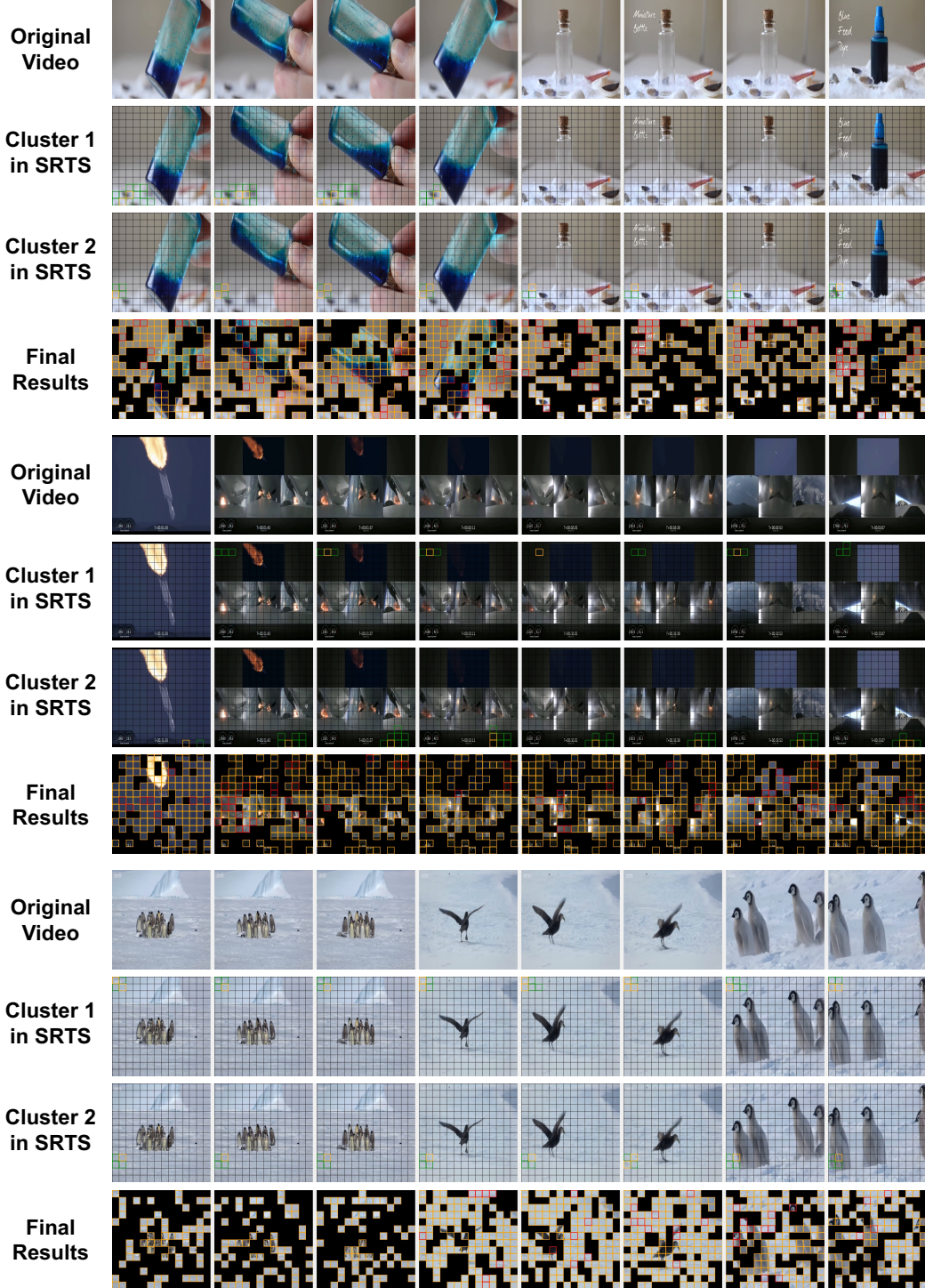


Figure 4: Visualization of the ST-SimDiff process. Token selection is decomposed into the communities recovered by SRTS (shown as grids in Cluster 1 and Cluster 2) and the event-driven tokens flagged by DETS. The final result renders the synergy explicit: sparse representative tokens (yellow) encode stable content, whereas dense event tokens (red) concentrate on dynamic actions.

To render the behavior of ST-SimDiff concrete, Figure 4 presents intermediate and final token selections on a representative video sequence depicting a dynamic object-manipulation task. Consider first the Similarity-based Representative Token Selection (SRTS): the rows labeled “Cluster 1” and “Cluster 2” expose how the graph community detection algorithm operates, grouping spatially and temporally redundant regions—the static background and stationary objects—into cohesive communities. The figure thereby confirms that SRTS identifies stable visual elements and compresses them to a handful of central “representative tokens,” visible as the sparse yellow boxes in the final results, which is precisely what enables the model to absorb extensive background redundancy at negligible token cost.

The Difference-based Event Token Selection (DETS) supplies the complementary behavior at dynamic moments. In the “Final Results” row, the tokens triggered by large temporal difference are marked with dense red bounding boxes that coincide with the moving hand and the shifting blue liquid, confirming that the method localizes exactly the “turning points” at which the temporal difference crosses the threshold τ_{diff} . This mechanism preserves fine-grained action detail and rapid visual change, allowing the model to capture the essence of the event without being overwhelmed by static content.

The overlay in the final results renders the synergy of the two components directly legible: sparse yellow tokens hold the stable background while dense red tokens concentrate on the rapid motion. This contrast constitutes direct visual evidence for our core motivation. By sampling densely through rapid changes, ST-SimDiff avoids the blind spot of similarity-only approaches; by pruning spatio-temporal redundancy, it recovers the compression that importance-based methods leave unrealized. The two behaviors together strike the intended balance between computational efficiency and the preservation of critical visual information.

B COMPARISON RESULTS ON QWEN2.5-VL

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
Qwen2.5-VL-64	74.2	62.7	51.8	62.9	54.2	64.8	59.2	63.0
Qwen2.5-VL-128	77.4	65.8	56.1	66.4	56.2	65.1	60.4	63.9
Token Retain Ratio $r = 30\%$								
ST-SimDiff-64	73.8	61.7	51.8	62.4	52.2	62.1	56.8	62.7
ST-SimDiff-128	75.4	64.7	55.2	65.1	53.1	65.0	58.6	64.4
Token Retain Ratio $r = 50\%$								
ST-SimDiff-64	74.2	62.1	52.2	62.9	54.2	63.2	58.4	62.7
ST-SimDiff-128	76.8	66.3	55.8	66.3	54.9	65.3	59.8	64.5

Table 4: Generalization to a third architecture: ST-SimDiff applied to Qwen2.5-VL-7B with 64 and 128 input frames (%), on VideoMME, LongVideoBench, and EgoSchema at token retain ratios of $r = 30\%$ and $r = 50\%$. The full-token model (Upper Bound) is listed for reference; at $r = 50\%$ the compressed model recovers essentially the full-token accuracy, and on VideoMME (64 frames) matches it exactly.

To probe the generality of our framework beyond the two base models of the main paper, we apply ST-SimDiff to the recently released Qwen2.5-VL model and evaluate on VideoMME and LongVideoBench, thereby testing whether its efficacy transfers to a third, independently developed architecture rather than being tied to any particular design.

The results in Table 4 confirm that it does. At a 30% token retain ratio, ST-SimDiff (64 frames) scores 62.4 on VideoMME and 56.8 on LongVideoBench; raising the budget to 50% lifts these to 62.9 and 58.4, respectively. Most telling, at the 50% ratio the VideoMME score matches the full-token upper bound exactly (62.9), while the LongVideoBench score remains closely comparable to its full-performance counterpart. That a training-free reduction to half the tokens recovers full VideoMME accuracy and near-full LongVideoBench accuracy on a third architecture indicates that

the framework’s benefits are not specific to any single base model, but transfer across independently developed designs.

C ABLATION STUDY RESULTS ON NVILA

Method	VideoMME	LongVideoBench	EgoSchema	
Token Retain Ratio $r = 30\%$				
Baseline	59.3	53.8	50.2	
+ Sim	Spatial	59.5	54.5	50.9
	Temporal	59.6	54.3	51.2
	Spa. + Tem.	59.9	54.6	51.3
++ Diff (Proposed)	60.2	55.2	51.7	
Token Retain Ratio $r = 50\%$				
Baseline	60.9	55.3	51.5	
+ Sim	Spatial	61.1	55.7	51.9
	Temporal	61.3	55.6	51.8
	Spa. + Tem.	61.4	56.0	52.3
++ Diff (Proposed)	61.7	56.5	52.5	

Table 5: Component ablation on the NVILA-Video-8B model (%), on VideoMME, LongVideoBench, and EgoSchema at token retain ratios of $r = 30\%$ and $r = 50\%$. As in the main paper, the Baseline is progressively augmented with the similarity module (+Sim; spatial, temporal, and joint variants) and then the difference module (++Diff); the complete framework is in **bold**. The monotone improvements, and the dominance of the joint spatio-temporal variant, mirror those observed on LLaVA-Video, indicating that the contribution of each component is not architecture-specific.

To verify that each component generalizes across base models, we repeat the component ablation on the NVILA-Video-8B model (Liu et al., 2024c), following exactly the protocol of the main-paper study: beginning from a Baseline of fundamental importance-based pruning, we add the similarity selection module (+ Sim) and then the difference selection module (++ Diff).

As Table 5 shows, the results reaffirm the framework’s design and track the LLaVA-Video-7B trends closely. Introducing the similarity module (+ Sim) improves accuracy steadily at both the 30% and the 50% retain ratio. Among its three variants—spatial-only (Spatial), temporal-only (Temporal), and joint spatio-temporal (Spa. + Tem.)—the joint variant is again the strongest: at $r = 50\%$ it reaches 61.4% on VideoMME, exceeding both single-axis variants and reconfirming that jointly modeling spatio-temporal association is the more effective route to identifying redundancy. Building on the joint similarity module, adding the difference selection module (++ Diff) to form the complete ST-SimDiff framework produces the largest single gain in the table. The consistency of this pattern with the LLaVA-Video results shows that both core modules—similarity and difference selection—generalize across architectures, and reaffirms our central idea of handling redundancy and change in concert.

D ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON LLaVA-VIDEO

The similarity threshold τ_{sim} and the difference threshold τ_{diff} are the two key hyperparameters of our framework, governing the selection of representative and transitional tokens respectively. Figure 6 reports how performance responds as each is swept. Both exhibit the same rise-then-fall profile, with a well-defined optimum and a broad surrounding range over which accuracy remains essentially flat. The behavior admits a direct interpretation: for τ_{sim} , too low a value yields imprecise community detection, whereas too high a value fractures the integrity of semantic clusters; for τ_{diff} , too low a value admits noise through over-sensitivity, whereas too high a value causes key events

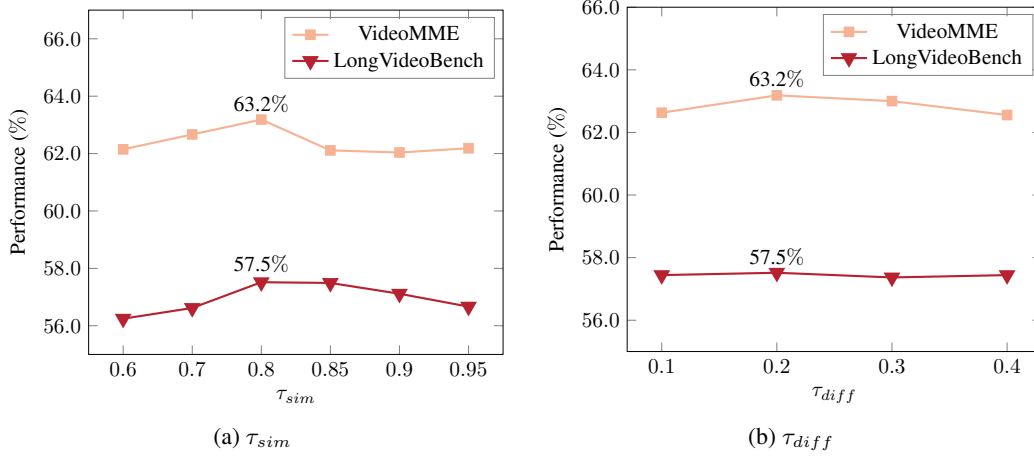


Figure 5: Sensitivity of ST-SimDiff to the similarity threshold τ_{sim} and the difference threshold τ_{diff} on LLaVA-Video, evaluated on VideoMME and LongVideoBench. Performance rises and then falls around each optimum yet varies only slightly across the entire sweep, indicating that the method is robust to both hyperparameters rather than finely tuned to a single value.

to be missed. Performance peaks at $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$, and we adopt this configuration throughout the remaining experiments.

E ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON NVILA-VIDEO

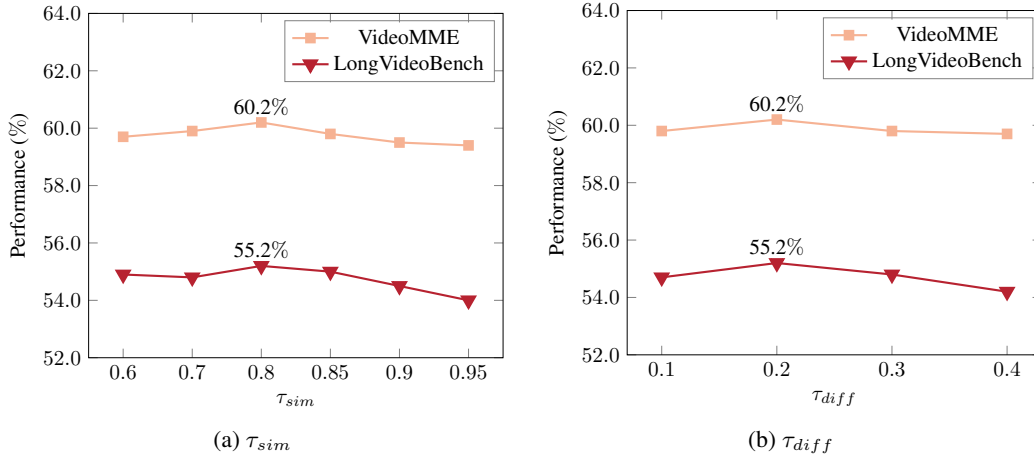


Figure 6: Sensitivity of ST-SimDiff to the similarity threshold τ_{sim} and the difference threshold τ_{diff} on NVILA-Video-8B, evaluated on VideoMME and LongVideoBench. The same rise-then-fall profile and broad flat region observed on LLaVA-Video recur here, confirming that the optimal thresholds transfer across base models.

To establish that the robustness of our hyperparameters is not an artifact of a single architecture, we repeat the threshold sensitivity study on the NVILA-Video-8B model. Fig. 6 charts the effect of the similarity threshold τ_{sim} and the difference threshold τ_{diff} on VideoMME and LongVideoBench.

Impact of Similarity Threshold τ_{sim} . Sweeping τ_{sim} over $[0.6, 0.95]$ leaves performance largely unaffected. Both benchmarks peak at $\tau_{sim} = 0.8$ —60.2% on VideoMME and 55.2% on LongVideoBench—and although accuracy dips slightly when the threshold is set too high or too low, the overall fluctuation is minimal, which confirms that the method is stable across a wide band of similarity thresholds.

918 **Impact of Difference Threshold τ_{diff} .** Sweeping τ_{diff} over $[0.1, 0.4]$ exhibits the same strong ro-
919 bustness. Both benchmarks are optimal near $\tau_{\text{diff}} = 0.2$ —60.2% on VideoMME and 55.2% on
920 LongVideoBench—indicating that a difference threshold near 0.2 most effectively captures key
921 events while compressing tokens.

922 Taken together, these studies confirm that ST-SimDiff is robust to its key hyperparameters and,
923 crucially, that the optimal setting $\tau_{\text{sim}} = 0.8$, $\tau_{\text{diff}} = 0.2$ transfers across benchmarks and base
924 models—which is precisely why we adopt it uniformly in the main experiments.
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